

Material	Category	Code	Paragraph
<i>Flexible</i> (continued) M Paper	multiwall multiwall, water-resistant	13M1 13M2	6.5.5.2
H Rigid plastics	for solids, filled or discharged by gravity, fitted with structural equipment for solids, filled or discharged by gravity, freestanding for solids, filled or discharged under pressure, fitted with structural equipment for solids, filled or discharged under pressure, freestanding for liquids, fitted with structural equipment for liquids, freestanding	11H1 11H2 21H1 21H2 31H1 31H2	6.5.5.3
HZ Composite with plastics inner receptacle*	for solids, filled or discharged by gravity, with rigid plastics inner receptacle for solids, filled or discharged by gravity, with flexible plastics inner receptacle for solids, filled or discharged under pressure, with rigid plastics inner receptacle for solids, filled or discharged under pressure, with flexible plastics inner receptacle for liquids, with rigid plastics inner receptacle for liquids, with flexible plastics inner receptacle	11HZ1 11HZ2 21HZ1 21HZ2 31HZ1 31HZ2	6.5.5.4
G Fibreboard	for solids, filled or discharged by gravity	11G	6.5.5.5
<i>Wooden</i> C Natural wood	for solids, filled or discharged by gravity, with inner liner	11C	6.5.5.6
D Plywood	for solids, filled or discharged by gravity, with inner liner	11D	
F Reconstituted wood	for solids, filled or discharged by gravity, with inner liner	11F	

The code shall be completed by replacing the letter 'Z' by a capital letter in accordance with 6.5.1.4.1.2 to indicate the nature of the material used for the outer packaging.

- 6.5.1.4.4 The letter 'W' may follow the IBC code. The letter 'W' signifies that the IBC, although of the same type as indicated by the code, is manufactured to a specification different from those in section 6.5.3 and is considered equivalent in accordance with the provisions in 6.5.1.1.2.

6.5.2 Marking

6.5.2.1 Primary marking

- 6.5.2.1.1 Each IBC manufactured and intended for use according to these provisions shall bear durable marks which are legible and placed in a location so as to be readily visible. Letters, numbers and symbols shall be at least 12 mm high and shall show:

- .1 the United Nations packaging symbol:



This symbol shall not be used for any purpose other than certifying that a packaging, a flexible bulk container, a portable tank or a MEGC complies with the relevant requirements in chapters 6.1, 6.2, 6.3, 6.5, 6.6, 6.7 or 6.9. For metal IBCs on which the mark is stamped or embossed, the capital letters "UN" may be applied instead of the symbol;

- .2 the code designating the type of IBC according to 6.5.1.4;
.3 a capital letter designating the packing group(s) for which the design type has been approved:
 "X" for packing groups I, II and III (IBCs for solids only);
 "Y" for packing groups II and III; or
 "Z" for packing group III only;
.4 the month and year (last two digits) of manufacture;
.5 the State authorizing the allocation of the marks, indicated by the distinguishing sign used on vehicles in international road traffic;

- .6 the name or symbol of the manufacturer and other identifications of the IBC as specified by the competent authority;
- .7 the stacking test load in kilograms. For IBCs not designed for stacking, the figure "0" shall be shown;
- .8 the maximum permissible gross mass in kilograms.

The primary marks required above shall be applied in the sequence of the subparagraphs above. The marks required by 6.5.2.2 and any further mark authorized by a competent authority shall still enable the primary marks to be correctly identified.

Each mark applied in accordance with .1 to .8 and with 6.5.2.2 shall be clearly separated, e.g. by a slash or space, so as to be easily identifiable.

6.5.2.1.2 IBCs manufactured from recycled plastics material as defined in 1.2.1 shall be marked "REC". For rigid IBCs, this mark shall be placed near the marks prescribed in 6.5.2.1.1. For the inner receptacle of composite IBCs, this mark shall be placed near the marks prescribed in 6.5.2.2.4.

6.5.2.1.3 Examples of marks for various types of IBCs in accordance with .1 to .8 above:



11A/Y/02 99/
NL/...* 007/
5500/1500

For a metal IBC for solids discharged by gravity and made from steel/ for packing groups II and III/ manufactured in February 1999/ authorized by the Netherlands/ manufactured by . . . *(name of manufacturer) and of a design type to which the competent authority has allocated serial number 007/ the stacking test load in kilograms/ and the maximum permissible gross mass in kilograms.



13H3/Z/03 01/
F/...* 1713/
0/1500

For a flexible IBC for solids discharged by gravity and made from woven plastics with a liner/ not designed to be stacked.



31H1/Y/04 99/
GB/...* 9099/
10800/1200

For a rigid plastics IBC for liquids made from plastics with structural equipment withstanding the stack load.



31HA1/Y/05 01/
D/...* 1683/
10800/1200

For a composite IBC for liquids with a rigid plastics inner receptacle and steel outer packaging.



11C/X/01 02/
S/...* 9876/
3000/910

For a wooden IBC for solids with an inner liner and authorized for packing group I solids.



11G/Z/06 02/
I/...* 962/
0/500

For a fibreboard IBC/ not designed to be stacked.



11D/Y/07 02/
E/...* 261/
3240/600

For a plywood IBC with inner liner.

6.5.2.1.4 Where an IBC conforms to one or more than one tested IBC design type, including one or more than one tested packaging or large packaging design type, the IBC may bear more than one mark to indicate the relevant performance test requirements that have been met. Where more than one mark appears on an IBC, the marks shall appear in close proximity to one another and each mark shall appear in its entirety.

6.5.2.2 Additional marking

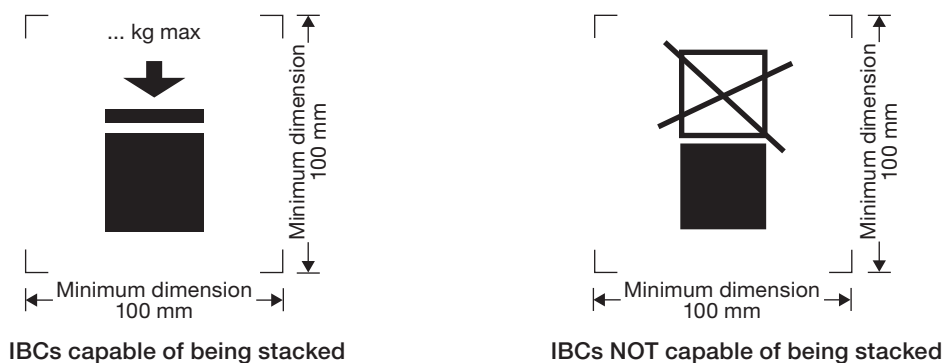
6.5.2.2.1 Each IBC shall bear the marks required in 6.5.2.1 and, in addition, the following information, which may appear on a corrosion-resistant plate permanently attached in a place readily accessible for inspection:

Note: For metal IBCs, this plate shall be a corrosion-resistant metal plate.

Additional marks	Category of IBC				
	Metal	Rigid plastics	Composite	Fibreboard	Wooden
Capacity in litres ^a at 20°C	X	X	X		
Tare mass in kg ^a	X	X	X	X	X
Test (gauge) pressure, in kPa or bar, ^a if applicable		X	X		
Maximum filling/discharge pressure in kPa or bar, ^a if applicable	X	X	X		
Body material and its minimum thickness in mm	X				
Date of last leakproofness test, if applicable (month and year)	X	X	X		
Date of last inspection (month and year)	X	X	X		
Serial number of the manufacturer	X				

The unit used shall be indicated.

6.5.2.2.2 The maximum permitted stacking load applicable shall be displayed on a symbol as shown in the figures below. The symbol shall be durable and clearly visible.



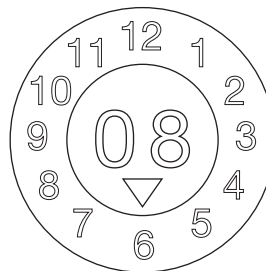
The minimum dimensions shall be 100 mm x 100 mm. The letters and numbers indicating the mass shall be at least 12 mm high. The area within the printer's marks indicated by the dimensional arrows shall be square. Where dimensions are not specified, all features shall be in approximate proportion to those shown. The mass marked above the symbol shall not exceed the load imposed during the design type test (see 6.5.6.6.4) divided by 1.8.

Note: The provisions of 6.5.2.2.2 shall apply to all IBCs manufactured, repaired or remanufactured as from 1 January 2011. The provisions of 6.5.2.2.2 of the IMDG Code (amendment 36-12) may continue to be applied to all IBCs manufactured, repaired or remanufactured between 1 January 2011 and 31 December 2016.

6.5.2.2.3 Each flexible IBC may also bear a pictogram or pictograms indicating the recommended lifting methods.

6.5.2.2.4 Inner receptacles that are of composite IBC design type shall be identified by the application of the marks indicated in 6.5.2.1.1.2, .3, .4 where this date is that of the manufacture of the plastics inner receptacle, .5 and .6. The UN packaging symbol shall not be applied. The marks shall be applied in the sequence shown in 6.5.2.1.1. They shall be durable, legible and placed in a location so as to be readily accessible for inspection after assembling the inner receptacle in the outer casing. When the marks on the inner receptacle are not readily accessible for inspection due to the design of the outer casing, a duplicate of the required marks on the inner receptacle shall be placed on the outer casing preceded by the wording "Inner receptacle". This duplicate shall be durable, legible and placed in a location so as to be readily accessible for inspection.

The date of the manufacture of the plastics inner receptacle may alternatively be marked on the inner receptacle adjacent to the remainder of the marks. In such a case, the date may be waived from the remainder of the marks. An example of an appropriate marking method is:



Note 1: Other methods that provide the minimum required information in a durable, visible and legible form are also acceptable.

Note 2: The date of manufacture of the inner receptacle may be different from the marked date of manufacture (see 6.5.2.1), repair (see 6.5.4.5.3) or remanufacture (see 6.5.2.4) of the composite IBC.

6.5.2.2.5 Where a composite IBC is designed in such a manner that the outer packaging is intended to be dismantled for transport when empty (such as for return of the IBC for re-use to the original consignor), each of the parts intended to be detached when so dismantled shall be marked with the month and year of manufacture and the name or symbol of the manufacturer and other identification of the IBC as specified by the competent authority (see 6.5.2.1.1.6).

6.5.2.3 Conformity to design type

The marks indicate that the IBCs correspond to a successfully tested design type and that the provisions referred to in the certificate have been met.

6.5.2.4 Marking of remanufactured composite IBCs (31HZ1)

The marks specified in 6.5.2.1.1 and 6.5.2.2 shall be removed from the original IBC or made permanently illegible and new marks shall be applied to an IBC remanufactured in accordance with these provisions of this Code.

6.5.3 Construction requirements

6.5.3.1 General requirements

6.5.3.1.1 IBCs shall be resistant to or adequately protected from deterioration due to the external environment.

6.5.3.1.2 IBCs shall be so constructed and closed that none of the contents can escape under normal conditions of transport, including the effects of vibration, or by changes in temperature, humidity or pressure.

6.5.3.1.3 IBCs and their closures shall be constructed of materials compatible with their contents, or be protected internally, so that they are not liable:

- .1 to be attacked by the contents so as to make their use dangerous;
- .2 to cause the contents to react or decompose, or form harmful or dangerous compounds with the IBCs.

6.5.3.1.4 Gaskets, where used, shall be made of materials not subject to attack by the contents of an IBC.

6.5.3.1.5 All service equipment shall be so positioned or protected as to minimize the risk of escape of the contents owing to damage during handling and transport.

6.5.3.1.6 IBCs, their attachments and their service and structural equipment shall be designed to withstand, without loss of contents, the internal pressure of the contents and the stresses of normal handling and transport. IBCs intended for stacking shall be designed for stacking. Any lifting or securing features of IBCs shall be of sufficient strength to withstand the normal conditions of handling and transport without gross distortion or failure and shall be so positioned that no undue stress is caused in any part of the IBC.

6.5.3.1.7 Where an IBC consists of a body within a framework, it shall be so constructed that:

- .1 the body does not chafe or rub against the framework so as to cause material damage to the body,
- .2 the body is retained within the framework at all times,
- .3 the items of equipment are fixed in such a way that they cannot be damaged if the connections between body and frame allow relative expansion or movement.

- 6.5.3.1.8 Where a bottom discharge valve is fitted, it shall be capable of being made secure in the closed position and the whole discharge system shall be suitably protected from damage. Valves having lever closures shall be able to be secured against accidental opening and the open or closed position shall be readily apparent. For IBCs containing liquids, a secondary means of sealing the discharge aperture shall also be provided, such as by a blank flange or equivalent device.

6.5.4 Testing, certification and inspection

6.5.4.1 Quality assurance

IBCs shall be manufactured, remanufactured, repaired and tested under a quality assurance programme which satisfies the competent authority, in order to ensure that each manufactured, remanufactured or repaired IBC meets the provisions of this chapter.

Note: ISO 16106:2020, *Transport packages for dangerous goods – Dangerous goods packagings, intermediate bulk containers (IBCs) and large packagings – Guidelines for the application of ISO 9001*, provides acceptable guidance on procedures which may be followed.

6.5.4.2 Test provisions

IBCs shall be subjected to design type tests and, if applicable, to initial and periodic inspections and tests in accordance with 6.5.4.4.

6.5.4.3 Certification

In respect of each design type of IBC, a certificate and mark (as in 6.5.2) shall be issued attesting that the design type, including its equipment, meets the test provisions.

6.5.4.4 Inspection and testing

Note: See also 6.5.4.5 for tests and inspections on repaired IBCs.

6.5.4.4.1 Every metal, rigid plastics and composite IBC shall be inspected to the satisfaction of the competent authority:

- .1 before it is put into service (including after remanufactured), and thereafter at intervals not exceeding five years, with regard to:

- .1 conformity to the design type, including marks;
- .2 internal and external condition; and
- .3 proper functioning of service equipment.

Thermal insulation, if any, need be removed only to the extent necessary for a proper examination of the body of the IBC.

- .2 at intervals of not more than two and a half years with regard to:

- .1 external condition; and
- .2 proper functioning of service equipment.

Thermal insulation, if any, need be removed only to the extent necessary for a proper examination of the body of the IBC.

Each IBC shall correspond in all respects to its design type.

6.5.4.4.2 Every metal, rigid plastics and composite IBC for liquids, or for solids which are filled or discharged under pressure, shall undergo a suitable leakproofness test. This test is part of a quality assurance programme as stipulated in 6.5.4.1 which shows the capability of meeting the appropriate test level indicated in 6.5.6.7.3:

- (a) before it is first used for transport;
- (b) at intervals of not more than two and a half years.

For this test the IBC shall be fitted with the primary bottom closure. The inner receptacle of a composite IBC may be tested without the outer casing, provided the test results are not affected.

6.5.4.4.3 A report of each inspection and test shall be kept by the owner of the IBC at least until the next inspection or test. The report shall include the results of the inspection and test and shall identify the party performing the inspection and test (see also the marking requirements in 6.5.2.2.1).

6.5.4.4.4 The competent authority may at any time require proof, by tests in accordance with this chapter, that the IBCs meet the provisions of the design type tests.